1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In TMPRSS2-ERG–positive prostate tumors that harbor KDM6A mutations, AR-driven transcription is reprogrammed toward a stromal remodeling gene program, resulting in an increased stromal-to-tumor ratio (TSR). This elevated TSR creates physical and transcriptional barriers that impair interaction and penetration of Lu-177–based radiopharmaceuticals (e.g., Lu-177–PSMA), leading to reduced therapeutic efficacy. Therefore, within the TMPRSS2-ERG+/KDM6A-mutant subset, higher TSR predicts poorer Lu-177 therapy response, independent of baseline receptor (PSMA) expression levels. Mechanistically, KDM6A loss may upregulate ECM/fibroblast-activating pathways (e.g., collagen deposition, matrix remodeling, perivascular niche alteration) and reinforce AR-driven networks that sustain stromal remodeling. Predictions include: (1) higher TSR correlates with lower Lu-177 uptake and diminished tumor control; (2) pharmacologic or genetic modulation of KDM6A or AR signaling that reduces TSR improves Lu-177 delivery and efficacy; (3) stromal-targeting or anti-fibrotic interventions that decrease TSR enhance Lu-177 radiopharmaceutical response in TMPRSS2-ERG+/KDM6A-mutant tumors. The hypothesis remains testable by integrating isogenic models, imaging-based TSR assessment, radiopharmaceutical uptake metrics, and therapeutic outcome measurements.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TMPRSS2-ERG, KDM6A
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMPRSS2-ERG: TMPRSS2-ERG fusion status does not uniformly predict sensitivity to Lu-177–based radiopharmaceuticals; what molecular or microenvironmental features in fusion-positive tumors determine therapy response? 2) Tier 1 path used: the single path links TMPRSS2-ERG fusion → androgen/AR signaling → prostate cancer → 177Lu (therapy context) and connects BM18 subtype mutated in KDM6A; I built on TMPRSS2-ERG and KDM6A because the path implicates a fusion-driven, AR-regulated program modified by KDM6A epigenetic state.

KDM6A: KDM6A) and AR-driven transcription alter expression of receptor programs and stromal remodeling, which in turn can affect intratumoral delivery/target availability for Lu-177 agents.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	AR	Supporting modifier from knowledge graph

Secondary	ERG	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- AR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ERG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.